

NEPTUNE MARITIME INDUSTRIES PTE. LTD.

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FACTORY ACCEPTANCE TEST (FAT) REPORT

INTEGRATED BRIDGE NAVIGATION SYSTEM

MV PACIFIC HORIZON — IMO No. 9876543

Document Number	NMI-FAT-2025-0847
Revision	02 — Final
Report Date	22 October 2025
FAT Dates	14 October 2025 to 16 October 2025
Vessel Name	MV PACIFIC HORIZON
IMO Number	9876543
Vessel Type	Bulk Carrier — 75,000 DWT (Handymax)
Flag State	Republic of the Marshall Islands
Shipowner / Operator	Pacific Bulk Carriers Holding Ltd., Hong Kong
Shipyard	Hyundai Heavy Industries Co., Ltd. — Ulsan, South Korea
Hull Number	HHI-2025-H3127
Classification Society	Lloyd's Register of Shipping (LR)
Equipment Supplier	Neptune Maritime Industries Pte. Ltd., Singapore
Contract Reference	PBC-NMI-2025-CON-041 dated 15 March 2025
Overall FAT Result	PASS — APPROVED FOR SHIPMENT

SIGNATURES — PREPARED, REVIEWED AND ACCEPTED BY:

ROLE	NAME / COMPANY / SIGNATURE	DATE SIGNED
Prepared By (FAT Engineer)	Chua Wei Ming — Senior Test Engineer Neptune Maritime Industries Pte. Ltd. _____	22 Oct 2025
Reviewed By (QC Manager)	Lim Boon Kiat — Quality Control Manager Neptune Maritime Industries Pte. Ltd. _____	22 Oct 2025
Witnessed By (LR Surveyor)	Mr. James O'Connor — LR Surveyor Lloyd's Register — Survey No. LR/SG/2025/0312 _____	16 Oct 2025

ROLE	NAME / COMPANY / SIGNATURE	DATE SIGNED
Owner's Representative	Capt. (Ret.) Zhang Wei — Technical Superintendent Pacific Bulk Carriers Holding Ltd. _____	16 Oct 2025
Flag State Inspector	Mr. David Lee — RMIS Authorized Inspector Republic of Marshall Islands Ship Registry _____	16 Oct 2025

1. INTRODUCTION AND SCOPE OF TEST

1.1 Purpose of This Report

This Factory Acceptance Test (FAT) Report documents the testing, verification, and formal acceptance of the Integrated Bridge Navigation System (IBNS) supplied by Neptune Maritime Industries Pte. Ltd. (the 'Supplier') for fitment aboard MV PACIFIC HORIZON, a 75,000 DWT Handymax bulk carrier under construction at Hyundai Heavy Industries, Ulsan Shipyard, South Korea.

The FAT was conducted at the Supplier's facility at 9 Tuas Basin Link, Singapore 638783, from 14 to 16 October 2025. The purpose is to verify, in a controlled factory environment, that all navigation equipment supplied under Contract PBC-NMI-2025-CON-041 complies with agreed technical specifications, international conventions, IMO performance standards, classification society rules, and flag state requirements prior to shipment.

This report is prepared in accordance with Lloyd's Register Rules for the Classification of Ships (2025 Edition), IMO Resolution A.694(17), and project-specific FAT Procedure Document NMI-FAT-PROC-2025-0847.

1.2 Background and Vessel Particulars

MV PACIFIC HORIZON is being constructed for Pacific Bulk Carriers Holding Ltd. of Hong Kong. The vessel will trade worldwide carrying dry bulk commodities including iron ore, coal, grain, and fertilizers. As a vessel engaged in international voyages exceeding 3,000 GT, it is subject to the full requirements of SOLAS Chapter V (Safety of Navigation), SOLAS Chapter IV (Radiocommunications), and applicable IMO performance standards for shipborne navigational and radio communication equipment.

1.3 Applicable Regulations and Standards

Reference	Title / Description	Applicable Equipment
SOLAS V/Reg 19	Carriage requirements for shipborne navigational systems and equipment	All navigation equipment
IMO Res. A.694(17)	General Requirements for Shipborne Radio Equipment forming Part of GMDSS	GMDSS, VHF, MF/HF
IMO Res. MSC.192(79)	Revised Performance Standards for Radar Equipment	X-Band & S-Band Radar
IMO Res. MSC.232(82)	Revised Performance Standards for ECDIS	ECDIS
IMO Res. MSC.74(69) Annex 3	Recommendations on Performance Standards for AIS	AIS Class A Transponder
IMO Res. MSC.116(73)	Performance Standards for Marine Transmitting Heading Devices (THDs)	Gyrocompass / FOG
IEC 60945 Ed.4	Maritime Navigation and Radiocommunication Equipment — General Requirements	All equipment
IEC 62388 Ed.2	Maritime Navigation and Radiocommunication Equipment — Shipborne Radar	X-Band & S-Band Radar
IEC 62923 Ed.1	Bridge Alert Management (BAM) — Shipborne Systems	BAM / Integrated Bridge
LR Rules Pt.5, Ch.2	Lloyd's Register Rules for Electrical and Electronic Equipment (2025)	All equipment

2. PERSONNEL AND ATTENDANCE

2.1 FAT Team Composition

The following personnel were present and actively participated in the FAT. All Neptune Maritime Industries test personnel hold valid competency certificates in the testing and commissioning of marine electronic systems. The Lloyd's Register Surveyor is an independent third-party witness. The Owner's Representative is authorized by Pacific Bulk Carriers Holding Ltd. to accept or reject equipment. The Flag State Representative from the Republic of Marshall Islands Ship Registry witnessed all SOLAS-mandatory tests.

#	Name	Organization	Role	Days Present
1	Chua Wei Ming	Neptune Maritime Industries	Lead FAT Engineer	14–16 Oct
2	Tan Ah Kow	Neptune Maritime Industries	Navigation Systems Technician	14–16 Oct
3	Ravi Subramaniam	Neptune Maritime Industries	GMDSS / Comms Specialist	14–16 Oct
4	Park Ji-Young	Neptune Maritime Industries	Software Integration Engineer	14–15 Oct
5	Lim Boon Kiat	Neptune Maritime Industries (QC)	QC Manager / Reviewer	16 Oct
6	Mr. James O'Connor	Lloyd's Register, Singapore	Class Surveyor (Witness)	14–16 Oct
7	Capt. (Ret.) Zhang Wei	Pacific Bulk Carriers Holding Ltd.	Owner's Technical Representative	14–16 Oct
8	Ms. Kim Su-Yeon	Hyundai Heavy Industries	Shipyard Representative	14–16 Oct
9	Mr. David Lee	RMIS Flag State Authority	Flag State Authorized Inspector	15–16 Oct
10	Mr. Anders Eriksson	Furuno Electric Co., Ltd.	OEM Technical Representative (Radar)	14–15 Oct

3. EQUIPMENT SCHEDULE AND IDENTIFICATION

The following table identifies all equipment items within the scope of this FAT, together with serial numbers, software/firmware versions, and type approval references. Serial numbers were verified by physical inspection against equipment nameplates and Supplier's records. The LR Surveyor independently verified serial numbers on all SOLAS-mandatory equipment.

#	Equipment Description	Manufacturer / Model	Serial Number	SW/FW	Type Approval	Result
1	X-Band Radar (Primary) 25 kW, IEC 62388	Furuno FAR-3230-BB	FAR3230-25-0841	SW 3.11.02	MED/4/0.2/0065	PASS
2	S-Band Radar (Secondary) 30 kW, IEC 62388	Furuno FAR-3230S-BB	FAR3230S-30-0842	SW 3.11.02	MED/4/0.2/0066	PASS
3	ECDIS (Primary) IMS S-63/S-57	JRC JAN-9201	JAN9201-P-0318	SW 4.02.01	MED/4/0.1/0210	PASS
4	ECDIS (Backup) IMS S-63/S-57	JRC JAN-9201	JAN9201-B-0319	SW 4.02.01	MED/4/0.1/0210	PASS
5	Fiber Optic Gyrocompass (FOG) — Master	Anschutz SmartPilot III d	SP3D-FOG-8812	FW 2.07	MED/4/0.3/0011	PASS
6	Gyrocompass Repeater (3 units)	Anschutz SmartPilot III d	SP3D-REP-0041/2/3	FW 2.07	—	PASS
7	AIS Class A Transponder	Saab R4 Transponder	R4AIS-2025-1092	SW 2.15.00	MED/4/0.2/0091	PASS
8	GNSS / GPS Navigator (Primary)	Furuno GP-170	GP170-P-04411	SW 1.51	MED/4/0.3/0112	PASS
9	GNSS / GPS Navigator (Backup)	Furuno GP-170	GP170-B-04412	SW 1.51	MED/4/0.3/0112	PASS
10	Speed Log (Dual-Axis EM Log)	JRC JLN-900	JLN900-2025-0081	FW 3.01	MED/4/0.3/0048	PASS
11	Echo Sounder (Single Beam)	JRC JFE-380	JFE380-2025-0099	FW 2.14	MED/4/0.3/0018	PASS
12	VHF DSC Radio (Bridge) x2	Sailor SP3520-D	SP3520-0814/0815	SW 3.21	MED/4/0.4/0071	PASS
13	INMARSAT-C (GMDSS) + EGC	Sailor 6110 Mini-C	S6110-2025-0293	SW 5.02.01	MED/4/0.4/0022	PASS
14	MF/HF SSB Radio + DSC	Sailor SP3525 MF/HF	SP3525-2025-0401	SW 2.10	MED/4/0.4/0101	PASS
15	SART Radar Transponder x2	McMurdo S20 SART	S20-0081/0082	—	MED/4/0.4/0033	PASS
16	AIS-SART MOB Device x2	Ocean Signal rescueME AIS1	AIS1-1104/1105	—	MED/4/0.4/0177	PASS
17	NAVTEX Receiver	Furuno NX-700A	NX700A-2025-0302	SW 2.01	MED/4/0.4/0088	PASS
18	Bridge Alert Management (BAM)	Neptune Maritime NAVIGATOR-X	NAVX-2025-0088	SW 1.08.03	LR/TA/2024/1198	PASS
19	Voyage Data Recorder (VDR)	Furuno VR-7000	VR7000-2025-0518	SW 3.09.00	MED/4/0.1/0193	PASS
20	Autopilot System	Anschutz NautiPilot 5000	NP5000-2025-0209	SW 4.11	MED/4/0.2/0015	PASS

4. X-BAND AND S-BAND RADAR SYSTEMS

4.1 Equipment Description and Technical Specifications

The vessel is fitted with a primary 25 kW X-Band radar (9.3 GHz) and a secondary 30 kW S-Band radar (3.0 GHz), both manufactured by Furuno Electric Co., Ltd. Both systems carry EU MED Wheelmark type approval and comply with IMO Resolution MSC.192(79) and IEC 62388:2013 Ed.2. The X-Band radar is the primary system used for close-range target detection, collision avoidance, and port approach navigation. The S-Band provides superior performance in precipitation and sea clutter. Both radars are equipped with ARPA functionality capable of auto-acquiring and tracking a minimum of 100 targets simultaneously with continuous CPA/TCPA computation and collision alarms.

Parameter	X-Band Radar (FAR-3230-BB)	S-Band Radar (FAR-3230S-BB)
Frequency Band	9.3 GHz (X-Band, 3 cm)	3.0 GHz (S-Band, 10 cm)
Peak Transmit Power	25 kW	30 kW
Antenna Rotation Speed	24 / 42 RPM (selectable)	24 / 42 RPM (selectable)
Antenna Beam Width (Horizontal)	1.23 deg @ 6.5 ft antenna	2.25 deg @ 8 ft antenna
Maximum Range	96 nautical miles	96 nautical miles
Minimum Range	Less than 25 m	Less than 25 m
Range Accuracy	Less than 1% of range scale or 30 m	Less than 1% of range scale or 30 m
Bearing Accuracy	Within +/-1 deg	Within +/-1 deg
ARPA Target Capacity	100 auto + 20 manual	100 auto + 20 manual
Display Size	19-inch SXGA LCD	19-inch SXGA LCD
Operating Temperature	-15 deg C to +55 deg C	-15 deg C to +55 deg C

4.2 Visual and Physical Inspection

Inspection Item	Observations / Findings	Ref. Clause	Result
Antenna Scanner (X-Band)	Scanner rotates freely. No physical damage. All fasteners secure. Cable entry sealed. Excellent condition.	IEC 62388 §4.1	PASS
Antenna Scanner (S-Band)	Scanner rotates freely. No physical damage. Bearing condition good. Cable entry sealed.	IEC 62388 §4.1	PASS
Transceiver Unit (X-Band)	All panels intact. No dents or corrosion. Ventilation grilles clear. Magnetron reads 000002 hrs.	IEC 62388 §4.2	PASS
Transceiver Unit (S-Band)	All panels intact. Minor cosmetic scratch on lower panel — noted and accepted by Owner's Rep (see NCR-001). Magnetron reads 000004 hrs.	IEC 62388 §4.2	PASS
Display Units (both)	Both displays free of scratches. Daylight readable. All controls operational. Labels legible.	IEC 62388 §5.1	PASS
Nameplate Verification	All nameplates legible. Serial numbers match FAT documentation exactly.	LR Rules	PASS
Cable and Connector Inspection	All coaxial antenna cables intact, correctly terminated, watertight glands fitted. No chafing.	IEC 60945 §9	PASS
Earth Bonding	Earth bonding cables fitted to all units, correctly sized. Continuity confirmed with test meter.	IEC 60945 §9	PASS

4.3 Performance Tests — X-Band Radar (Furuno FAR-3230-BB)

Performance testing was conducted using calibrated test equipment including a Radar Performance Monitor (RPM) and signal generator. Tests were performed to verify compliance with IMO MSC.192(79) and IEC 62388 Ed.2. The Furuno OEM representative Mr. Anders Eriksson was present throughout and confirmed manufacturer-specific test parameters.

Test Description	Acceptance Criterion	Measured Result	Clause	Result
Transmitter Power Output	22 kW minimum (spec 25 kW)	24.7 kW measured	IEC 62388 §7.3	PASS
Minimum Range (Target Detection)	Less than 25 m with 10 m2 RCS target	15 m measured	MSC.192 §2.3	PASS
Range Accuracy (3 nm scale)	1% or 30 m (whichever greater)	±18 m at 2.5 nm	IEC 62388 §7.8	PASS
Bearing Accuracy	Within ±1 deg of true bearing	±0.5 deg measured (0.9 deg worst case)	IEC 62388 §7.9	PASS
Range Discrimination	30 m or less between two targets	22 m measured	MSC.192 §2.5	PASS
Bearing Discrimination	2.5 deg or less between two targets	1.8 deg measured	MSC.192 §2.5	PASS
Sea Clutter (STC) Control	Suppresses clutter; targets retained	Sea clutter suppressed; test target at 0.3 nm retained	IEC 62388 §9.5	PASS
Rain Clutter (FTC) Control	Suppresses rain clutter; targets retained	Rain clutter suppressed; 1 nm target retained	IEC 62388 §9.5	PASS
North-Up Stabilized Mode	Heading data accepted; display rotates correctly	N-Up mode confirmed with FOG heading input	MSC.192 §5.2	PASS
True Motion Mode	Position tracking with speed/heading input	TM mode confirmed; own ship trail correct	MSC.192 §5.2	PASS
VRM Accuracy (two VRMs)	1% full scale accuracy	VRM1: ±0.05 nm, VRM2: ±0.04 nm	IEC 62388 §9.10	PASS
EBL Accuracy (two EBLs)	0.5 deg accuracy	EBL1: ±0.3 deg, EBL2: ±0.4 deg	IEC 62388 §9.10	PASS
Parallel Index Lines	PI lines operational	PI lines displayed and offset correctly	MSC.191 §7	PASS

4.4 ARPA Target Tracking Tests

ARPA functionality was tested using the Furuno integrated ARPA simulator mode, which injects synthetic target data into the radar display at known positions, speeds, and courses. This allows verification of CPA/TCPA computation accuracy, alarm functions, and tracking performance. The ARPA simulator is approved for FAT testing by Furuno Electric and was verified against OEM reference data.

ARPA Test Description	Acceptance Criterion	Test Result	Clause	Result
Target Acquisition — Auto Mode	100 auto targets; acquisition within 3 sweeps	100 targets acquired; avg. acquisition in 2.1 sweeps	MSC.192 §4.2	PASS
Target Acquisition — Manual Mode	20 manual targets in addition to auto	20 manual targets confirmed alongside 100 auto	MSC.192 §4.2	PASS

ARPA Test Description	Acceptance Criterion	Test Result	Clause	Result
CPA / TCPA Computation Accuracy	CPA error 0.3 nm or less; TCPA 0.5 min after steady state	CPA error: 0.15 nm; TCPA: 0.2 min after 3 min	MSC.192 §4.5	PASS
Collision Alarm (CPA/TCPA)	Audible and visual alarm at threshold	Alarm at CPA 0.5 nm / TCPA 6 min — both alarms confirmed	MSC.192 §4.6	PASS
Lost Target Alarm	Alarm within 3 sweeps of target loss	Lost target alarm at 2 sweeps	MSC.192 §4.7	PASS
Vector Display (True / Relative)	Correct vector in both modes	Both modes confirmed; correct agreement with course and speed	MSC.192 §4.4	PASS
History Trail (3 and 6 min)	Trail in both 3-min and 6-min settings	Trails displayed correctly in both settings	MSC.192 §4.4	PASS
Guardzones (Two Independent)	Alarm on target entry into guardzone	Both guardzones triggered correctly with simulated target entry	MSC.192 §4.9	PASS
Target Recovery After Manoeuvre	Re-tracks within 1 min after 90-deg turn	Recovery confirmed at 41 seconds after simulated 90-deg turn	MSC.192 §4.8	PASS

5. ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEM (ECDIS)

5.1 Equipment Description

Two ECDIS units are supplied — primary and backup — both JRC JAN-9201 units with software version 4.02.01. Both carry EU MED Wheelmark type approval MED/4/0.1/0.0210 and comply with IMO Resolution MSC.232(82) and IHO publications S-52, S-57 Ed.3.1, and S-63 Ed.1.2. The ECDIS units serve as the primary means of navigation replacing mandatory paper charts per SOLAS Reg. V/19.2.10.

Each ECDIS is connected to primary and backup GPS/GNSS sensors, the FOG gyrocompass, the dual-axis speed log, and the AIS transponder via IEC 61162-1 (NMEA 0183) interfaces. The backup ECDIS operates entirely independently from the primary, maintaining its own sensor connections to prevent common-mode failure. This independence was specifically tested and witnessed by the LR Surveyor.

5.2 ECDIS Performance and Functional Tests

Test Description	Acceptance Criterion	Observed Result	Clause	Result
ENC Display — Base Display	Correct base elements per S-52 displayed	Coastlines, safe depth contour, isolated dangers, prohibited areas all displayed correctly	S-52 §3	PASS
ENC Display — Standard Display	Additional chart elements per S-52 standard	Standard display confirmed with all additional layers active	S-52 §3	PASS
Safe Contour Setting	Configurable safety contour; unsafe depths highlighted	Safety contour set to 10 m; unsafe areas highlighted in blue. Danger highlight correct.	MSC.232 §6.1	PASS
Own Ship Position Display	GPS position continuously updated on chart	Position updating every 1 second from primary GPS. XTD visible on chart.	MSC.232 §5.1	PASS
GPS/GNSS Sensor Switchover	Backup GPS auto-selects if primary fails; alarm generated	Primary GPS disconnected — backup selected within 3 sec; alarm generated. Confirmed by LR Surveyor.	MSC.232 §10.2	PASS
Route Planning Function	Waypoints, track, distance, ETA computed correctly	Route planned Singapore to Rotterdam (18 waypoints); distance, ETE and ETA correct vs. manual calculation.	MSC.232 §5.3	PASS
Route Monitoring — Anti-Grounding	Anti-grounding alarm at configurable look-ahead distance	Alarm triggered at 3 nm look-ahead; hazard highlighted on chart.	MSC.232 §9.3	PASS
Waypoint Approach Alarm	Alarm at configurable distance before waypoint	Alarm at 1.0 nm before waypoint as configured; audible and visual alarms confirmed.	MSC.232 §9.3	PASS
XTD (Cross-Track Distance) Alarm	Alarm when vessel deviates beyond set XTD limit	XTD alarm at 0.2 nm deviation as set. Alert passed to BAM system.	MSC.232 §9.3	PASS
Speed/Heading from FOG and Log	Heading from gyro and speed from log displayed	Heading and speed displayed correctly; confirmed match with sensor readings.	MSC.232 §5.2	PASS
AIS Targets on ECDIS	AIS targets overlaid with correct symbology	AIS targets displayed with correct IHO S-52 symbology; target data popup correct.	MSC.232 §5.4	PASS
Radar Overlay on ECDIS	Radar echo overlay functional	Radar overlay from X-Band radar displayed correctly; matches chart features.	MSC.232 §5.5	PASS

Test Description	Acceptance Criterion	Observed Result	Clause	Result
S-63 Chart Permit Verification	Valid permits accepted; expired permits rejected	Valid permits accepted; expired permit correctly rejected with error message.	S-63 §4	PASS
Chart Update Application	ENC updates applied; update records maintained	Test update applied; history log correct.	S-52 §8	PASS
Night Mode / Dusk Mode Display	Display brightness adjustable; night colour scheme operational	All display modes (Day, Dusk, Night) correct; minimum brightness meets requirements.	IEC 60945 §6	PASS
Backup ECDIS Independence Test	Backup ECDIS operates independently of primary	Primary ECDIS powered off; Backup continued normal operation using own sensor inputs without interruption. LR Surveyor confirmed.	MSC.232 §3.3	PASS

6. GYROCOMPASS, GNSS AND AIS SENSORS

6.1 Fiber Optic Gyrocompass (FOG) — Anschutz SmartPilot IIId

The Anschutz SmartPilot IIId uses interferometric fiber optic technology to measure angular rate and derive heading. Unlike conventional mechanical gyrocompasses, the FOG contains no rotating mechanical parts, eliminating lubrication requirements and significantly reducing maintenance demands. The system achieves heading accuracy of ± 0.1 deg (1 sigma) in static conditions and better than ± 0.5 deg under dynamic vessel motion. It feeds heading data to all connected bridge equipment via NMEA 0183 HDT/HDG sentences at 10 Hz and provides an analog synchro output for three repeater compass units at the conning position, port bridge wing, and starboard bridge wing.

Test Description	Acceptance Criterion	Measured Result	Ref.	Result
Start-Up Time from Cold Start	Heading output within 5 minutes	Heading available at 3 min 42 sec (cold start)	MSC.116(73) §3	PASS
Heading Accuracy — Static	± 0.5 deg (1 sigma) corrected for latitude	± 0.08 deg (1 sigma) over 10 minutes static	MSC.116(73) §4.1	PASS
Heading Repeatability	± 0.25 deg or less over 30-minute period	± 0.06 deg measured over 30 minutes — excellent stability	MSC.116(73) §4.2	PASS
Rate of Turn Output	ROT within ± 1 deg/min accuracy	ROT output correct; ± 0.4 deg/min accuracy confirmed	IEC 62882 §5	PASS
Synchro Output (Repeater Drive)	3 repeaters within ± 0.5 deg of master	All 3 repeaters confirmed within ± 0.3 deg of master	MSC.116(73) §5	PASS
NMEA HDT/HDG Sentence Output	Correct sentences at 10 Hz update rate	HDT and HDG confirmed at 10 Hz; data correct	IEC 61162-1	PASS
Power Supply Interruption Recovery	Heading reacquired within 60 seconds	Heading reacquired at 58 sec after 30-sec power interruption	MSC.116(73) §6	PASS

6.2 GNSS / GPS Navigation System — Furuno GP-170

Two independent GNSS receiver units are installed — Primary GPS and Backup GPS. Both units receive signals from GPS (US), GLONASS (Russia), and SBAS constellations. The antennas are positioned at separate locations on the vessel to eliminate the possibility of a single obstruction affecting both receivers. Each unit outputs position data in NMEA 0183 GGA, GLL, RMC, VTG, and ZDA sentences to all connected navigation equipment.

Test Description	Acceptance Criterion	Measured Result	Ref.	Result
Position Accuracy (open sky, survey point)	CEP 10 m (GPS only); 5 m (SBAS)	CEP 2.3 m (SBAS-aided, 100 measurements over 30 min)	IMO Res. A.915	PASS
NMEA GGA Sentence Output	GGA at 1 Hz; position correct	GGA at 1 Hz; position confirmed against DGPS reference	IEC 61162-1	PASS
NMEA ZDA Sentence Output	ZDA (time/date) at 1 Hz	UTC time correct; date confirmed	IEC 61162-1	PASS
Primary to Backup Switchover	Auto-switchover within 5 seconds; alarm generated	Switchover at 2.8 seconds; alarm confirmed on DDU and ECDIS	MSC.232 §10.2	PASS
Position Jump Alarm	Alarm if position changes more than 1 nm instantaneously	Alarm triggered correctly when simulated 2 nm position jump injected	MSC.232 §10.3	PASS

6.3 AIS Class A Transponder — Saab R4

The Saab R4 AIS Class A Transponder complies with IMO Resolution MSC.74(69) Annex 3, ITU-R M.1371-5, and IEC 62287-1. The system automatically broadcasts vessel static data (MMSI: 538009187, IMO: 9876543, Name: PACIFIC HORIZON, Type: 70 Cargo, Dims: 229x32 m), dynamic data (position, SOG, COG, heading, ROT, navigational status), and voyage-related data at intervals per SOTDMA protocol.

Test Description	Acceptance Criterion	Observed Result	Ref.	Result
Static Data Programming and Display	All static data correctly programmed and transmitted	MMSI, IMO, Name, Type, Dims all confirmed correct in AIS Message Type 1	IEC 62287-1 §6	PASS
Dynamic Data Transmission Rate	Reporting rate per ITU M.1371 (10s at speed above 14 kts)	10-second rate confirmed at simulated 15 kt speed	ITU M.1371 §3.3	PASS
Position Accuracy in AIS Message	Position from GNSS within AIS; RAIM status indicated	GPS position in Msg Type 1 confirmed correct; RAIM status = 1 (normal)	IEC 62287-1 §8.3	PASS
Heading from Gyrocompass in AIS	True heading from FOG transmitted in AIS message	Heading 045.0 deg (simulated) confirmed in AIS Message Type 1	IEC 62287-1 §8.3	PASS
Target Display — Receive Mode	AIS targets received and displayed on MKD and ECDIS	3 simulated AIS targets confirmed on MKD and ECDIS with correct symbols	IEC 62288 §5	PASS
Silent Mode (VHF Off)	AIS silenced per Master's authority; alarm generated	Silent mode activated; alarm generated on BAM system	SOLAS V/19.2.4	PASS
Channel Management	CH 87B and 88B default; regional channel change accepted	Default channels confirmed; channel change via MKD tested successfully	ITU M.1371 §4.2	PASS

7. GLOBAL MARITIME DISTRESS AND SAFETY SYSTEM (GMDSS)

7.1 GMDSS Equipment Suite Overview

MV PACIFIC HORIZON operates worldwide in GMDSS Sea Areas A1, A2, A3, and A4 (polar areas). The vessel carries a full GMDSS equipment suite as mandated by SOLAS Chapter IV. The EPIRB (406 MHz) is supplied under a separate contract and is excluded from this FAT scope.

#	Equipment	GMDSS Function	Sea Area	SOLAS Req.	Result
1	VHF DSC Controller x2 (Sailor SP3520-D)	Distress alerting, Ship-to-ship communications	A1	SOLAS IV/7.1.1	PASS
2	MF/HF SSB + DSC (Sailor SP3525)	Distress alerting, Long range communications	A2, A3, A4	SOLAS IV/9.1	PASS
3	INMARSAT-C + EGC (Sailor 6110 Mini-C)	Ship-earth station, MSI reception	A3, A4	SOLAS IV/10.1.1	PASS
4	NAVTEX Receiver (Furuno NX-700A)	MSI — NavWarnings, weather, SAR	A1, A2	SOLAS IV/7.1.2	PASS
5	SART Radar Transponder x2 (McMurdo S20)	Location of survival craft by radar	All areas	SOLAS IV/7.1.3	PASS
6	AIS-SART MOB Device x2 (Ocean Signal AIS1)	Location of person in distress via AIS	All areas	MSC.1/Circ.1515	PASS
7	406 MHz EPIRB x1 (separate supply)	Satellite distress alerting	All areas	SOLAS IV/7.1.6	N/A

7.2 VHF DSC Radio Tests — Sailor SP3520-D

Test Description	Acceptance Criterion	Observed Result	Ref.	Result
DSC Distress Alert Transmission (Internal)	DSC alert generated on CH70; correct MMSI transmitted	Alert in internal test mode; MMSI correct; self-check passed	SOLAS IV/7.1.1	PASS
Transmit Power — 25W and 1W	25W and 1W selectable; within ±30% of rated	25W: 23.1W measured; 1W: 1.05W measured — both within tolerance	IEC 60945 §8	PASS
Receiver Sensitivity CH16	Less than -107 dBm for 12 dB SINAD	-108.5 dBm measured — satisfactory	ITU-R M.489	PASS
Interoperability — Unit 1 to Unit 2	Units communicate on CH16 and working channels	Clear communications on CH16 and CH06 in conference call test	SOLAS IV	PASS
Scanning Function (Priority Scan)	CH16 in priority scan; alarm on CH16 call	Priority scan confirmed; audible alarm on CH16 call from Unit 2	IEC 62287-3	PASS
Remote Handset (Bridge Wings)	Handsets at both bridge wings operational	Both port and starboard handsets operational on both VHF units	MSC.1/Circ.	PASS

8. VOYAGE DATA RECORDER (VDR) AND BRIDGE ALERT MANAGEMENT

8.1 Voyage Data Recorder — Furuno VR-7000

The Furuno VR-7000 is a full VDR complying with IMO Resolution MSC.333(90) and IEC 61996-1 Ed.2. Though a Simplified VDR (S-VDR) would be permissible for this vessel type per SOLAS V/20, the Owner has specified a full VDR for maximum recording capability. Data recorded includes bridge audio (all microphones), radar video from both radars, ECDIS screenshots, AIS data, GPS position and time, heading, speed, depth, wind, rudder angle, engine telegraph, thruster status, watertight and fire door status, and VHF radio communications. All data is stored in a 12-hour rolling loop on a certified float-free capsule rated to 6,000 m depth.

Test Description	Acceptance Criterion	Observed Result	Clause	Result
Data Input — GPS Position (GGA)	Position recorded at 1-second interval or less	1-second recording confirmed by playback	IEC 61996-1 §7.5	PASS
Data Input — Heading (HDT)	Heading at 1-second interval or less	1-second recording; heading from FOG correct	IEC 61996-1 §7.5	PASS
Data Input — Speed (VBW/VHW)	Speed from speed log recorded	STW from EM log recorded correctly; playback confirmed	IEC 61996-1 §7.5	PASS
Data Input — Radar Video (X-Band)	Radar image recorded every 15 seconds	Image at 15-sec intervals confirmed; playback correct	IEC 61996-1 §7.6	PASS
Data Input — AIS NMEA Data	AIS messages recorded	VDM/VDO sentences recorded; playback shows correct targets	IEC 61996-1 §7.5	PASS
Bridge Audio Recording	All bridge mics recorded; voice intelligible	4 microphone channels confirmed; voice intelligible on playback	IEC 61996-1 §7.4	PASS
VHF Communications Recording	VHF radio audio from both units recorded	Both VHF channels confirmed; playback intelligible	IEC 61996-1 §7.4	PASS
Capsule Status and Self-Check	VDR passes self-check; no fault indications	Self-check completed without fault; float-free mechanism operational	IEC 61996-1 §9	PASS
Time Synchronisation	VDR time synchronised to GPS UTC	VDR clock synchronised to GPS UTC; offset 0.0 seconds	IEC 61996-1 §7.5	PASS

8.2 Bridge Alert Management (BAM) — Neptune Maritime NAVIGATOR-X

The Neptune Maritime NAVIGATOR-X BAM System complies with IEC 62923-1 Ed.1 and IEC 62923-2 Ed.1. The BAM receives alert inputs from all connected bridge equipment and presents them on a unified alert management workstation. Alerts are prioritised into three categories (Alarm, Warning, Caution) and acknowledgement is managed in accordance with IEC 62923 requirements.

Test Description	Acceptance Criterion	Observed Result	Ref.	Result
Alert Reception from ECDIS (XTD Alarm)	ECDIS XTD alarm displayed on BAM	XTD alarm received and displayed on BAM within 1 second	IEC 62923-1 §4.2	PASS
Alert Reception from ARPA (CPA Alarm)	Radar CPA alarm received on BAM	CPA alarm displayed on BAM; alert category: Alarm	IEC 62923-1 §4.2	PASS
Alert Escalation (Unacknowledged)	Re-alerted after 30 sec if unacknowledged	Re-alert at 30-sec intervals until acknowledged; audible alarm re-triggered	IEC 62923-1 §5.3	PASS

Test Description	Acceptance Criterion	Observed Result	Ref.	Result
Acknowledgement Function	Alerts acknowledged from BAM workstation	Acknowledgement confirmed; alerts move to acknowledged list	IEC 62923-1 §5.4	PASS
Inhibit Function (Caution Class)	Caution-class alerts can be inhibited	Inhibit confirmed for caution-class only; alarm-class cannot be inhibited	IEC 62923-1 §5.5	PASS
Alert Transfer (Watchkeeping Handover)	Responsibility transferable between stations	Alert responsibility transferred; confirmed on both stations	IEC 62923-2 §4	PASS

9. SYSTEM INTEGRATION AND INTERFACE TESTS

Integration testing verifies correct data transfer between all bridge equipment via NMEA 0183 and proprietary data bus interfaces. All inter-equipment interfaces are routed through the Neptune Maritime Industries Data Distribution Unit (DDU) Model NMI-DDU-02, which provides signal isolation, buffering, and priority-based automatic switchover. Tests were conducted with all equipment powered simultaneously to replicate actual bridge operating conditions.

#	Data Source	Recipient	Data Type	Test Observation	Clause	Result
1	FOG Gyrocompass	ECDIS (Primary)	NMEA HDT @10Hz	Heading on ECDIS matches gyro exactly	IEC 61162-1	PASS
2	FOG Gyrocompass	X-Band Radar	NMEA HDT @10Hz	Radar N-Up mode correct; heading matches gyro	IEC 61162-1	PASS
3	FOG Gyrocompass	AIS Transponder	NMEA HDT @1Hz	AIS Msg 1 heading field matches gyro	IEC 61162-1	PASS
4	FOG Gyrocompass	Autopilot	NMEA HDT @10Hz	Autopilot receives heading; confirmed by display	IEC 61162-1	PASS
5	Primary GNSS	ECDIS (Primary)	NMEA GGA @1Hz	Position on ECDIS matches GPS; 1-sec update	IEC 61162-1	PASS
6	Primary GNSS	AIS Transponder	NMEA GGA @1Hz	AIS position = GPS position; confirmed correct	IEC 61162-1	PASS
7	Primary GNSS	VDR	NMEA GGA @1Hz	Position logged at 1-sec intervals; playback correct	IEC 61162-1	PASS
8	Speed Log (EM Log)	ECDIS (Primary)	NMEA VBW @1Hz	STW displayed on ECDIS; confirmed correct	IEC 61162-1	PASS
9	AIS Transponder	ECDIS (Primary)	NMEA VDM (var.)	AIS targets on ECDIS with correct symbology	IEC 61162-1	PASS
10	AIS Transponder	X-Band Radar	NMEA VDM (var.)	AIS targets in radar overlay; symbols correct	IEC 61162-1	PASS
11	ECDIS (Primary)	BAM System	IEC 62923 CAN	ECDIS alarms transmitted to BAM; all categories correct	IEC 62923	PASS
12	DDU Sensor Switchover	ECDIS/Radar/AIS	Automatic	Primary GPS and Gyro failure simulated; DDU switched to backup within 3 sec; alarm generated on all connected displays	NMI DDU Spec	PASS

10. DEVIATIONS, NON-CONFORMANCES, AND CORRECTIVE ACTIONS

During this FAT, two minor Non-Conformance Records (NCRs) were identified and formally documented. Both were classified as minor in nature and did not prevent the overall PASS status of the affected equipment items. Both NCRs were fully rectified during the FAT period prior to close-out of testing on Day 3. The LR Surveyor reviewed and accepted the corrective actions for both NCRs. Formal NCR documents are available in Annex E.

NCR	Date Found	Equipment	Description of Non-Conformance	Corrective Action Taken	Closed	Status
NCR-001	14 Oct 2025	S-Band Radar Transceiver	Minor cosmetic scratch approximately 40 mm in length observed on lower right panel of S-Band radar transceiver during visual inspection. Scratch is superficial and does not affect functionality, performance, or structural integrity. Ref: FAT Procedure §4.1.2	QC Manager confirmed scratch is cosmetic only, no penetration of protective coating. Anti-corrosion treatment applied. Scratch documented photographically and accepted in writing by Owner's Representative (Capt. Zhang Wei) on 15 October 2025. Accepted 'As-Is'.	15 Oct 2025	CLOSED
NCR-002	15 Oct 2025	ECDIS Primary JAN-9201	During Route Monitoring test, anti-grounding look-ahead time was displaying in UTC instead of Local Time (LT) as specified in Purchase Order Technical Specification Appendix B §3.2.11. This is a software configuration issue, not a type-approval non-compliance.	JRC Technical Support contacted. Configuration file updated to display look-ahead time in LT (user-configurable). Update applied to both Primary and Backup ECDIS units on 15 October 2025. Re-test performed on Day 3 (16 October 2025): PASS. Witnessed by LR Surveyor and Owner's Representative.	16 Oct 2025	CLOSED

11. TEST SUMMARY AND ACCEPTANCE STATEMENT

11.1 Equipment Test Summary

#	System / Equipment Group	No. of Tests	PASS	FAIL	N/A	Final Status
1	X-Band Radar — Visual Inspection	8	8	0	0	PASS
2	X-Band Radar — Performance Tests	13	13	0	0	PASS
3	X-Band and S-Band Radar — ARPA Tests	9	9	0	0	PASS
4	ECDIS (Primary and Backup) — Functional Tests	16	16	0	0	PASS
5	Gyrocompass (FOG) — Performance Tests	7	7	0	0	PASS
6	GNSS / GPS — Performance Tests	5	5	0	0	PASS
7	AIS Class A — Performance Tests	7	7	0	0	PASS
8	GMDSS — VHF DSC Tests	6	6	0	0	PASS
9	GMDSS — MF/HF, INMARSAT-C and NAVTEX Tests	9	9	0	0	PASS
10	VDR — Data Recording Tests	9	9	0	0	PASS
11	Bridge Alert Management — Integration Tests	6	6	0	0	PASS
12	System Integration — Interface Tests	12	12	0	0	PASS
13	SART and AIS-SART — Functional Tests	4	4	0	0	PASS
GRAND TOTAL		113	113	0	0	ALL PASS

11.2 Formal Acceptance Statement

On the basis of the Factory Acceptance Tests conducted at Neptune Maritime Industries Pte. Ltd., Singapore, from 14 to 16 October 2025, and following the satisfactory close-out of all Non-Conformance Records as detailed in Section 10, the undersigned parties confirm:

- All 20 equipment items have been tested and found to comply with applicable type approval standards, IMO performance standards, flag state requirements, and contractual technical specifications.
- A total of 113 individual test procedures were performed, all resulting in a PASS. Zero test failures remain outstanding.
- Two NCRs (NCR-001 and NCR-002) were raised during the FAT and formally closed prior to signing of this acceptance statement.
- The Lloyd's Register Surveyor has witnessed all SOLAS-mandatory tests and confirms the equipment meets LR Classification Rules and IMO requirements for installation aboard MV PACIFIC HORIZON.
- The RMIS Flag State Inspector confirms no objection to the equipment being installed on a Marshall Islands-flagged vessel.
- The equipment is hereby APPROVED FOR SHIPMENT to Hyundai Heavy Industries, Ulsan Shipyard, South Korea, for installation and commissioning aboard MV PACIFIC HORIZON.

ROLE	NAME / COMPANY / SIGNATURE	DATE SIGNED
Equipment Supplier (Neptune Maritime Industries)	Lim Boon Kiat — QC Manager Neptune Maritime Industries Pte. Ltd. _____	22 October 2025

ROLE	NAME / COMPANY / SIGNATURE	DATE SIGNED
Shipowner / Operator (Pacific Bulk Carriers)	Capt. (Ret.) Zhang Wei — Technical Superintendent Pacific Bulk Carriers Holding Ltd. _____	16 October 2025
Classification Society (Lloyd's Register)	Mr. James O'Connor — LR Surveyor Lloyd's Register — Survey No. LR/SG/2025/0312 _____	16 October 2025
Flag State Authority (Republic of Marshall Islands)	Mr. David Lee — RMIS Authorized Inspector Republic of Marshall Islands Ship Registry _____	16 October 2025
Shipyard (Hyundai Heavy Industries)	Ms. Kim Su-Yeon — Equipment Coordinator Hyundai Heavy Industries Co., Ltd. — Ulsan _____	16 October 2025

12. ANNEXES AND ATTACHMENTS

Annex	Title	Document Number	Status
Annex A	Pre-FAT Documentation Checklist and Sign-Off Sheet	NMI-FAT-DOC-2025-0847-A	Complete — Signed
Annex B	Type Approval Certificate Register — All Equipment	NMI-FAT-DOC-2025-0847-B	Complete — Current TACs Attached
Annex C	Calibration Certificates — All Test Instruments	NMI-FAT-DOC-2025-0847-C	Complete — Within Calibration Period
Annex D	Equipment Software / Firmware Version Record	NMI-FAT-DOC-2025-0847-D	Complete
Annex E	NCR Register — NCR-001 and NCR-002 with Close-Out Records	NMI-FAT-NCR-2025-0847	Complete — Both NCRs Closed
Annex F	Photographic Record of FAT — Day 1, 2 and 3	NMI-FAT-PHO-2025-0847	Complete — USB Media Attached
Annex G	NMEA Interface Matrix — Verified as-tested	NMI-FAT-ICD-2025-0847	Complete
Annex H	VDR Data Playback Verification Report	NMI-FAT-VDR-2025-0847	Complete — LR Surveyor Witnessed
Annex I	LR Surveyor's Attendance Certificate	LR/SG/2025/0312-ATT	Original Attached
Annex J	Owner's Acceptance Form (Signed Original)	PBC-NMI-FAT-ACC-2025	Original Attached

— END OF FACTORY ACCEPTANCE TEST REPORT —

Document No: NMI-FAT-2025-0847 | Revision: 02 | Date: 22 October 2025
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